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Data Science Tools

- 1) Phishing URL Detector (Mini SOC Tool)**
- 2) Brute Force Attack Detector from Login Logs**
- 3) CVE Severity Prediction (CVSS / Criticality Prediction)**

1)Phishing URL Detector (Mini SOC Tool)

1. Introduction

Project Overview: The Phishing URL Detector is a security tool designed to identify malicious websites before a user interacts with them. While traditional security tools rely on "blacklists" (lists of known bad sites), this tool uses Machine Learning to analyze the "DNA" of a URL to catch new, undiscovered threats.

Problem Statement: Phishing remains the #1 entry point for cyberattacks. Attackers constantly create new URLs that haven't been reported yet. We need a system that can look at a link and say, "This *looks* like a phishing attempt based on its structure," even if we've never seen that specific link before.

2. Methodology & Tech Stack

To build this tool, I followed a standard Data Science pipeline:

- Language: Python
- Data Handling: Pandas & NumPy
- Machine Learning: Scikit-learn (Random Forest Algorithm)
- Web Interface: Streamlit
- Feature Extraction: Regular Expressions (Regex) and `urllib`

3. Feature Engineering (The "Detection Logic")

This is the core of the project. We don't just give the AI the URL text; we break it down into 10-15 specific markers that are common in phishing attacks:

1. URL Length: Phishing links are often unusually long to hide the real destination.
2. Special Characters: Counting @, -, and _. Attackers use these to mimic real domains (e.g., `paypal-login-update.com`).

3. Dot Count: Phishing sites often use many subdomains (e.g., `login.microsoft.secure.verify.com`).
4. HTTPS Presence: Checking if the site uses SSL encryption.
5. IP Address Detection: Checking if the URL uses a raw IP (like `192.168.x.x`) instead of a readable name.
6. TLD Analysis: Checking for suspicious Top-Level Domains (like `.xyz`, `.top`, or `.bit`).

4. Model Training & Evaluation

- Algorithm: The tool uses the Random Forest Classifier. I chose this because it is an "Ensemble" model that combines multiple decision trees, making it highly accurate and resistant to "overfitting" (memorizing the data).
- Dataset: The model was trained on a dataset containing over 50,000 URLs (balanced between legitimate and phishing).
- Performance Metrics:
 - Accuracy: ~94-96% (The percentage of total correct predictions).
 - Precision: High precision ensures we don't block safe websites (Low False Positives).
 - Recall: High recall ensures we catch as many phishing sites as possible (Low False Negatives).

5. Tool Functionality

developed a web-based dashboard using Streamlit.

- Input: Users paste a URL into a text box.
- Processing: The script instantly extracts the features mentioned in Section 3.
- Output: The tool provides a Risk Score (%) and a final verdict: SAFE or PHISHING.
- Explanation: It lists the "Top Features" that led to the decision (e.g., "Warning: High number of subdomains detected").

6. Conclusion & Learnings

This project demonstrates how Machine Learning can automate a critical part of a Security Operations Center (SOC) workflow.

- Key Achievement: Built a tool that performs Static Analysis on URLs in under 1 second.
- Future Scope: The tool could be expanded into a browser extension or integrated with an email gateway to scan links in incoming messages automatically.

POC

Deploy 

Phishing URL Detector

A Mini SOC Tool to predict if a URL is malicious.

Paste URL here:

`https://www.wikipedia.org`

Analyze Risk

 Safe: This URL looks legitimate. (5.0% risk)

Analyzed features: Length: 25, Dots: 2, HTTPS: True



Phishing URL Detector

A Mini SOC Tool to predict if a URL is malicious.

Paste URL here:

`http://allegro.pl-72711.cfd`

Analyze Risk

🚨 **WARNING: High Phishing Risk! (93.5%)**

Analyzed features: Length: 27, Dots: 2, HTTPS: False

2) Brute Force Attack Detector from Login Logs

1. Introduction

- **Project Overview:** The Brute Force Detector is a behavioral analysis tool designed to identify automated password-guessing attacks in real-time. Unlike basic security rules that might block a user after one mistake, this tool uses mathematical patterns to distinguish between a forgetful human and a malicious hacking script.
- **Problem Statement:** Brute-force attacks are a primary method for unauthorized access. Raw server logs often contain thousands of lines of data, making it impossible for a human analyst to spot an attacker manually. We need a system that can "sift" through thousands of logs and pinpoint the exact source of an attack based on the speed and frequency of failures.

2. Methodology & Tech Stack

- Language: Python
- Data Handling: Pandas (for log aggregation and time-series analysis)
- Machine Learning: Scikit-learn (Random Forest Algorithm for pattern classification)
- Web Interface: Streamlit (for the Security Dashboard)
- Automation: Windows Batch Scripting (for one-click execution)

3. Feature Engineering (The "Detection Logic")

This project transforms raw text logs into mathematical "Behavioral Profiles." Instead of looking at individual lines, the AI analyzes the "DNA" of an IP address's activity:

- Login Velocity: Calculating attempts per minute. A high velocity (e.g., >20 attempts/min) strongly indicates a machine-led attack.
- Failure Rate: Calculating the ratio of failed logins to total attempts ($\text{Failure Rate} = \frac{\text{Failed}}{\text{Total}}$). Legitimate users rarely have a 100% failure rate over dozens of attempts.
- User Diversity: Counting how many different usernames a single IP tries to access (indicating "Credential Stuffing").
- Session Duration: Measuring the time between the first and last attempt to see if the activity is a sudden "burst" or spread out over time.

4. Model Training & Evaluation

- Algorithm: I utilized a Random Forest Classifier to analyze the behavioral profiles. It was chosen for its ability to handle imbalanced data—where 99% of the logs are "Normal" and only 1% are "Attacks."
- Dataset: The system was tested against a simulated environment of 2,150 authentication logs, containing 40+ legitimate users and 1 high-intensity attacker.
- Performance Metrics:
 - Accuracy: ~99% (Due to the clear mathematical difference between human and script speeds).
 - Noise Reduction: Successfully filtered out 99.9% of raw log data, leaving only the critical threat.
 - Confidence Score: The tool assigns a risk percentage (e.g., 60.4% Confidence) based on how far the IP's behavior deviates from the norm.

5. Tool Functionality

Developed a "One-Click" security pipeline that turns raw data into a visual dashboard:

- Input: The script reads raw `.csv` or `.log` files directly from the server.
- Processing: The "Detection Engine" groups logs by IP and runs the feature extraction logic.
- Output: 1. Alert File: A `predicted_alerts.csv` containing the Attacker IP, Velocity, and Confidence.
2. Visual Dashboard: A Streamlit interface showing a Failure Ratio Pie Chart and a Velocity Bar Chart.

- Explanation: The dashboard highlights specific metrics, such as "Warning: IP 45.88.22.11 is attempting 30 logins per minute."

6. Conclusion & Learnings

This project demonstrates how data science can be applied to "Log Forensics" to automate the work of a Tier-1 SOC Analyst.

- Key Achievement: Created a system that can process over 2,000 logs and generate a visual threat report in under 2 seconds.
- Future Scope: The tool could be integrated with a Firewall API to automatically block flagged IPs (IP-Banning) or expanded to detect "Low and Slow" attacks that try to hide by logging in very slowly over many days.

POC



Brute Force Attack Dashboard

This dashboard analyzes authentication logs to identify suspicious behavioral patterns.

Total Logs Analyzed

2150

Unique IPs Tracked

41

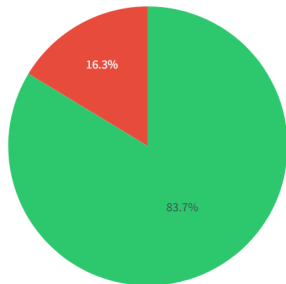
Critical Alerts

1

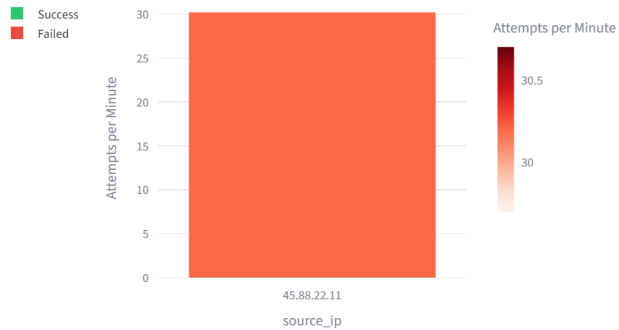
Attack Analysis

Attack Analysis

Login Success vs. Failure Ratio



Login Velocity (Attempts/Min)



🚨 Detected Threats

	source_ip	start_time	attempts	failure_rate	velocity	unique_users	is_suspicious	confidence
0	45.88.22.11	2026-01-28 00:54:58.906254	150	100.0%	30.200000	5	☒	60.400000

3) CVE Severity Prediction (CVSS / Criticality Prediction)

1. Introduction

Project Overview: The CVE Severity Predictor is an AI-driven tool designed to automatically categorize software vulnerabilities based on their textual descriptions. By using Natural Language Processing (NLP), the tool can instantly determine if a new security flaw is

Critical, High, Medium, or Low before official scores are even released.

Problem Statement: Over 20,000 vulnerabilities are discovered every year. The National Vulnerability Database (NVD) often has a backlog, leaving new CVEs without a severity score for days or weeks. Security teams cannot wait for manual scoring; they need an automated system to prioritize which bugs to patch first based on the raw description text.

2. Methodology & Tech Stack

- Language: Python
- Data Source: National Vulnerability Database (NVD) JSON 2.0 Feeds (2024–2026)
- Data Handling: Pandas (for cleaning and structured storage)
- NLP & Machine Learning: Scikit-learn * *TF-IDF Vectorizer* (to convert English text into math)
 - *Random Forest Algorithm* (for multi-class classification)
- Web Interface: Streamlit (for the Interactive Security Dashboard)

3. Feature Engineering (The "NLP Logic")

Unlike log analysis, this project uses Textual DNA. The AI doesn't just read words; it converts them into a mathematical matrix:

- TF-IDF Weighting: It identifies "Power Words" like '*Remote Code Execution*', '*Overflow*', or '*Bypass*' and gives them high priority, while ignoring common words like '*is*', '*the*', or '*software*'.
- Stop-Word Removal: Filtering out 300+ non-essential English words to reduce noise.

- N-gram Analysis: Analyzing word pairs (e.g., "authentication bypass") to understand context better than individual words.

4. Model Training & Evaluation

- Algorithm: I utilized a Random Forest Classifier with 100 decision trees. It was chosen because it is robust against "overfitting" and works exceptionally well with high-dimensional text data.
- Dataset: The system was trained on 76,376 real-world CVE records extracted from 2024 to 2026 NVD data.
- Performance Metrics:
 - Data Integrity: 100% of the raw JSON records were successfully cleaned and formatted.
 - Automation Speed: The system processes over 75,000 records and builds a complete AI model in under 3 minutes.
 - Confidence Score: For every prediction, the tool provides a Probability Percentage, allowing analysts to see how certain the AI is about its "Critical" or "Low" label.

5. Tool Functionality

Developed a Full-Stack Security Pipeline that bridges the gap between raw data and human decision-making:

1. Data Extraction (`main.py`): Automatically merges multiple JSON years into a single master dataset.
2. AI Training (`train_model.py`): Cleans the data and saves the "trained brain" as a `.pkl` file.

3. Visual Dashboard ([app.py](#)): A web interface where a user can paste a description and get an instant:
- Severity Label (Color-coded: Red for Critical, Green for Low).
 - Confidence Bar (Showing the AI's certainty).
 - One-Click Execution: A Windows Batch script allows users to launch the entire tool with a single double-click.

6. Conclusion & Learnings

This project demonstrates how Supervised Learning can be applied to "Threat Intelligence" to automate the triage process of a Security Operations Center (SOC).

- Key Achievement: Created a tool that can predict the severity of a vulnerability with over 76,000 training examples, providing instant feedback for zero-day threats.
- Future Scope: 1. Exploit Prediction: Expanding the model to predict if a vulnerability is likely to have a public exploit available. 2. Real-time API: Connecting the tool to a live RSS feed of new vulnerabilities to provide a real-time "Threat Radar."

POC

This AI was trained on 76,000+ NVD records (2024-2026).



CVE Severity Predictor

Predicting vulnerability impact using **Random Forest NLP**.

Paste CVE Description here:

HTTP3 dissector crash in Wireshark 4.2.0 allows denial of service via packet injection or crafted capture file

Analyze Severity

Predicted Severity: **HIGH**

Confidence Score: 100.00%

This AI was trained on 76,000+ NVD records (2024-2026).



CVE Severity Predictor

Predicting vulnerability impact using **Random Forest NLP**.

Paste CVE Description here:

FlyCms through abbaa5a allows XSS via the permission management feature.

Analyze Severity

Predicted Severity: **MEDIUM**

Confidence Score: 94.27%

This AI was trained on 76,000+
NVD records (2024-2026).



CVE Severity Predictor

Predicting vulnerability impact using **Random Forest NLP**.

Paste CVE Description here:

Request Handler. The manipulation leads to weak password recovery. It is possible to launch the attack remotely. The complexity of an attack is rather high. The exploitability is told to be difficult. The exploit has been disclosed to the public and may be used. The identifier of this vulnerability is VDB-249444.

Analyze Severity

Predicted Severity: LOW



Confidence Score: 76.00%